TECHNICAL DOCUMENTATION

Human Authenticity Analyzer Architecture & Tech Stack

A microservices-based facial intelligence platform for real-time and image-based analysis across emotion, age & gender, deepfake detection, and skin analysis.

6 Services
5 Python + 1 Next.js

4 Analysis Modules
Emotion · Age/Gender · Deepfake · Skin

Dual-Layer AI
Client-side + Server-side

SECTION 01

System Overview

🧠

ARCHITECTURE PATTERN

Microservices + Client-side AI

⚡

DETECTION MODES

Image Upload & Live Webcam

🔒

FACE GATING

Client-side before API call

🐍

BACKEND RUNTIME

Python · FastAPI · Uvicorn

⚛️

FRONTEND FRAMEWORK

Next.js 16 · React 19 · TypeScript

🤖

ML LIBRARIES

DeepFace · TensorFlow.js · OpenCV

SECTION 02

Architecture Diagram

```
┌──────────────────────────────────────────────────────────────────────┐ │ 👤 USER BROWSER
│ └──────────────────────┬───────────────────────────────────────────────┘ │
┌──────────────────────▼─────────────────────────────────────────────────┐ │ FRONTEND ·
Next.js 16 · React 19 · TypeScript · Port 3000 │ │ │ │ ┌─────────────────────┐
┌──────────────────────────────────┐ │ │ │ @vladmandic/ │ │ ApiClient.ts │ │ │ │ face-
api.js │ │ Centralized HTTP Client │ │ │ │ (TensorFlow.js) │ │ multipart/form-data →
services │ │ │ │ client-side only │ └─────────────────┬──────────────────┘ │ │ │ face gate
check │ │ │ │ └─────────────────────┘ │ HTTP POST │
└──────────────────────────────────────────┼─────────────────────────────┘ │
┌───────────────────────────────┼──────────────────────────┐ ▼ ▼ ▼ ▼ ▼ ┌───────────────┐
┌───────────────┐ ┌───────────────┐ ┌──────────────────┐ ┌───────────┐ │ Main API │ │ Age &
Gender │ │ Emotion │ │ Bruise/Marks │ │ Deepfake │ │ :8000 │ │ :8001 │ │ :8002 │ │ :8003 │
│ :8004 │ │ │ │ │ │ │ │ │ │ │ │ Skin+ │ │ DeepFace │ │ DeepFace │ │ OpenCV │ │ELA + FFT │
│ Deepfake │ │ FairFace │ │ FER/AffNet │ │ (YOLO stub) │ │OpenCV │ │ proxy │ │ UTKFace │ │
│ │ │ │ │ └───────────────┘ └───────────────┘ └──────────────────┘ └───────────────┘
└───────────┘
```

Note: Each Python service is fully independent with its own port, `requirements.txt`, and process. They share no state and can be started/stopped independently.

SECTION 03

End-to-End Request Flow

📤 Image Upload Mode

1 **User selects an image file**

File is read in the browser via the ImageUploader component.

2 **Client-side face gate (face-api.js)**

Before any network call, TensorFlow.js runs **SSD MobileNet** locally in the browser to confirm a face is present. If no face → error, no API call made.

```
detectFaceAndEmotions(img) → detections[]
```

3 **HTTP POST to Python microservice**

Image sent as `multipart/form-data` via `ApiClient.ts` to the relevant service endpoint.

```
POST localhost:800x/analyze-image
```

4 **Server-side ML inference**

FastAPI decodes image bytes with OpenCV, feeds into DeepFace or custom CV predictor, returns structured JSON result.

5 **Result stored → Results page**

JSON result saved to `sessionStorage`. User redirected to `/results/<module>` which reads from storage and renders charts/metrics.

📹 Live Webcam Mode (2-minute session)

1 **Camera stream opened**

WebcamCapture component accesses device camera via MediaDevices API.

2 **Face-api.js processes frames every 2 seconds**

Client-side inference loop runs `detectFaceAndEmotions()` at 2s intervals, streaming live age/gender/emotion data to the sidebar panel.

3 **Session timer counts down 120s**

SessionTimer component triggers `onComplete` callback after 2 minutes. Frame data is accumulated in a ref.

4

Session aggregated → results page

Accumulated age history and live demographics are averaged and stored. User redirected to results page showing trend charts.

SECTION 04

Microservices Breakdown

SERVICE	PORT	CORE LOGIC	ML / LIBRARY	ENDPOINTS
Main API	`:8000`	Skin tone + texture analysis, deepfake proxy	`OpenCV` `HSV` `Laplacian`	`/api/analyze-frame`
Age & Gender	`:8001`	Age regression + gender classification	`DeepFace` `FairFace` `UTKFace`	`/analyze-image` `/analyze-live-session`
Emotion	`:8002`	7-class emotion detection	`DeepFace` `FER2013` `AffectNet`	`/analyze-image` `/analyze-live-session`
Bruise/Marks	`:8003`	Bounding box detection of bruises, scars, marks	`OpenCV` `NumPy` `YOLO stub`	`/analyze-image` `/analyze-live-session`
Deepfake	`:8004`	Authenticity analysis via image forensics	`ELA` `FFT` `OpenCV`	`/analyze-image` `/analyze-live-session`
Frontend	`:3000`	UI, routing, live face detection, result rendering	`face-api.js` `TF.js` `Next.js`	Browser-only

SECTION 05

Full Tech Stack

FRONTEND

Framework	`Next.js 16 (App Router)`
Language	`TypeScript 5`
UI Library	`React 19`
Styling	`Tailwind CSS v4`
Animations	`Framer Motion 12`
Icons	`Lucide React`
Client AI	`@vladmandic/face-api v1.7`
State	`React Hooks + sessionStorage`
HTTP	`Fetch API (ApiClient.ts)`

BACKEND

Framework	`FastAPI`
Server	`Uvicorn (ASGI)`
Language	`Python 3.x`
Image Processing	`OpenCV (headless)`
Numerics	`NumPy`
Deep Learning	`DeepFace + tf-keras`
Uploads	`python-multipart`
CORS	`FastAPI CORSMiddleware`
Environment	`Python venv`

SECTION 06

Client-Side ML Models

Loaded from `/public/models` as static Next.js assets. All inference runs in-browser via TensorFlow.js — no server call required.

`ssdMobilenetv1`
Primary face detection with bounding boxes

`tinyFaceDetector`
Lightweight fallback face detector for speed

`faceLandmark68Net`
68-point facial landmark mapping

`faceRecognitionNet`
128-d face embedding extraction

`faceExpressionNet`
7-class emotion recognition (softmax)

`ageGenderNet`
Age regression + binary gender classification

Server-Side AI Techniques

SERVICE	TECHNIQUE	DETAILS
Emotion	Deep Neural Network	DeepFace → FER2013 / AffectNet CNN, 7 emotion classes
Age & Gender	Regression + Classification	DeepFace with FairFace, IMDB-WIKI, UTKFace pre-trained weights
Deepfake	ELA + FFT Forensics	JPEG recompression variance (ELA 60%) + high-frequency spectral energy (FFT 40%)
Skin Tone	Color Space Analysis	HSV V-channel average → Light/Medium/Dark classification
Skin Texture	Laplacian Variance	Edge variance → Clear / Moderate Texture / Visible Blemishes
Bruise/Marks	Object Detection (stub)	Placeholder for YOLO model; currently uses seeded NumPy mock

SECTION 07

Training Datasets

MODULE	DATASETS USED FOR TRAINING / FINE-TUNING
Emotion Detection	FER2013 · AffectNet · RAF-DB (Real-world Affective Faces Database)
Age & Gender	IMDB-WIKI · UTKFace · Adience
Deepfake Detection	FaceForensics++ · DFDC (DeepFake Detection Challenge Dataset)
Skin Tone & Clearness	Fitzpatrick17k · DDI (Diverse Dermatology Images) · ACNE04

SECTION 08

Key Architecture Patterns

Dual-Layer Face Detection

face-api.js performs a client-side face gate check *before* sending data to the backend. This prevents wasted API calls and improves UX by giving immediate feedback if no face is detected.

Session-Based Result Transfer

Results are written to `sessionStorage` after analysis. The `/results/` page reads from storage — decoupling the analysis step from the display step, enabling clean routing.

Microservice Independence

Each service is a self-contained FastAPI app with its own port and `requirements.txt`. Services share no state, can be developed/deployed independently, and route via `ApiClient.ts`.

Graceful Fallback Strategy

If DeepFace fails to import, each backend service automatically falls back to a seeded NumPy heuristic that produces deterministic mock results from pixel data — ensuring the UI never crashes.

Upload vs. Live Mode

Upload: Single image → backend inference → results page. **Live:** face-api.js processes webcam frames client-side every 2s for real-time stats. Frames are optionally batched to backend at session end.

Session Aggregation

Each backend service has an `aggregate_*_session()` function that averages multiple frame results — computing dominant values, confidence trends, and sampled timelines for chart display.

SECTION 09

Supported Fields & Beneficiaries

The Human Authenticity Analyzer is designed as a cross-domain platform. Its four analysis modules collectively serve the following professional fields and user groups.

🔐 Cybersecurity & Digital Forensics

The **Deepfake Detection** module directly supports digital forensic investigators, content moderation teams, and online trust & safety departments by identifying AI-generated or manipulated facial imagery using ELA and FFT spectral analysis.

Deepfake Detection | Content Moderation | Digital Forensics

🏥 Healthcare & Clinical Medicine

The **Bruise & Marks Detection** and **Skin Analysis** modules aid clinical staff, dermatologists, and triage practitioners in documenting visible physical injuries, tracking skin conditions, and assessing blemishes or discolorations remotely.

Bruise & Marks Detection | Dermatology | Skin Tone Analysis

⚖️ Law Enforcement & Legal

Law enforcement agencies and legal professionals can use the **Deepfake** and **Bruise Detection** modules for evidence validation — verifying authenticity of photographic evidence and documenting physical injury markers in abuse or assault cases.

Evidence Verification | Injury Documentation | Media Authentication

🧠 Psychology & Mental Health

The **Emotion Detection** module (7-class real-time recognition) supports psychologists, therapists, and mental health researchers in non-invasive affect monitoring, patient engagement analysis, and emotional state tracking during sessions.

Emotion Detection | Affect Monitoring | Behavioral Research

📊 Market Research & Demographics

Marketing analysts, UX researchers, and demographers benefit from the **Age & Gender Detection** module to gather real-time audience demographic data, measure emotional engagement with content, and profile user cohorts without requiring form-based input.

Age & Gender Detection | Audience Analytics | UX Research

🎓 Academic & AI Research

Researchers in computer vision, human-computer interaction, and AI fairness can use this platform as a benchmark tool or testbed — evaluating model accuracy, demographic bias, dataset generalization, and multi-modal fusion across all four analysis domains.

Computer Vision | AI Fairness | HCI Research

🛡️ Border Control & Identity Verification

Immigration authorities, airport security, and identity verification services can leverage the **Age & Gender** and **Deepfake Detection** modules to cross-verify submitted identity documents and detect presentation attacks or photo manipulations.

Identity Verification | Presentation Attack Detection | Age Verification

🎬 Media, Entertainment & EdTech

Media producers, game developers, and EdTech platforms use emotion and demographic analysis to personalize content delivery, measure learner engagement in real-time, adapt difficulty based on emotional state, and validate audience age-appropriateness.

Emotion Detection | Content Personalization | Learner Engagement

Cross-Domain Value: All four analysis modules (Emotion, Age & Gender, Deepfake, Skin/Marks) can operate independently or together in the All-in-One mode, making the platform adaptable to domain-specific workflows without requiring full system deployment.

SECTION 10

Research Gap Analysis

Despite the system's functional coverage across four facial analysis domains, several significant research gaps exist that limit its accuracy, fairness, and real-world applicability. These represent opportunities for future work.

G1

Demographic Bias & Dataset Imbalance

HIGH PRIORITY

The core training datasets (FER2013, IMDB-WIKI, UTKFace, Adience) are heavily skewed toward Western, lighter-skinned demographics. This results in measurably lower accuracy for South Asian, East Asian, and African subjects — particularly in age estimation and emotion recognition. No bias-correction or re-weighting technique is currently applied in the inference pipeline.

Impact: Undermines fairness across diverse global user bases; misclassification rates rise significantly for underrepresented groups.

G2

Deepfake Detection vs. Modern Diffusion Models

HIGH PRIORITY

The current deepfake detection relies on ELA (JPEG compression variance) and FFT (frequency domain energy) — heuristics designed to catch GAN-based artifacts. Modern deepfakes generated by Latent Diffusion Models (e.g., Stable Diffusion, DALL-E, Midjourney) produce photorealistic images that do not exhibit the same JPEG or high-frequency artifacts, rendering these methods largely ineffective. No neural-network-based deepfake classifier (e.g., trained on FaceForensics++) is integrated.

Impact: State-of-the-art AI-generated images may pass as "REAL" with high confidence, creating a critical false-negative risk.

G3

Bruise & Marks Detection — No Real ML Model

HIGH PRIORITY

The bruise and skin marks detection service currently uses a seeded NumPy random stub as a placeholder for a YOLO object detection model. No actual trained detector exists in the codebase. This means bounding boxes, labels, and confidence scores are entirely fabricated — making the service non-functional for real use. A fine-tuned YOLOv8 or Faster R-CNN model trained on annotated injury datasets (e.g., MIVIA, custom dermatology collections) is needed.

Impact: The bruise detection module produces no scientifically valid output, limiting applicability in healthcare and legal contexts entirely.

G4

No Explainability Layer (XAI)

MEDIUM PRIORITY

All four modules return prediction outputs (labels, scores, bounding boxes) without any explainability or interpretability mechanism. There is no Grad-CAM visualization, LIME explanation, or attention map indicating which facial regions drove each prediction. In high-stakes domains (healthcare, legal, security), decision transparency is critical for trust and auditability.

Impact: Results cannot be audited or contested; adoption in regulated industries (medical, legal) is blocked without interpretable outputs.

G5

Absence of Cross-Session Temporal Tracking

MEDIUM PRIORITY

Results are stored only in browser `sessionStorage`, which is cleared when the tab closes. There is no persistent database, user profile system, or longitudinal tracking across sessions. For clinical and research applications, tracking changes in emotion, age estimation, or skin condition over time (e.g., weekly) would provide substantially more value.

Impact: Prevents longitudinal studies, trend analysis, and any personalized or comparative use case across sessions.

G6

Single-Face Constraint & Occlusion Sensitivity

MEDIUM PRIORITY

The system is optimized for single, clearly visible, frontal-facing faces under adequate lighting. Multi-face scenes are not jointly analyzed. Occluded faces (masks, glasses, partial coverage), low-light environments, and extreme pose variations (profile, tilted) all significantly degrade detection accuracy without graceful degradation handling or pose-normalization preprocessing.

Impact: Real-world deployment environments (crowds, poor lighting, masked individuals) fall outside the reliable operating range.

G7

No Privacy, Consent & Ethics Framework

MEDIUM PRIORITY

The system collects and processes biometric facial data (a sensitive personal data category under GDPR, CCPA, and Sri Lanka's proposed Data Protection Act) without any consent management, anonymization pipeline, or on-device processing guarantee. There is no audit trail, data retention policy, or user rights mechanism (right to erasure, access, portability) implemented.

Impact: Significant legal and regulatory compliance risk; deployment in the EU or any jurisdiction with biometric data laws would require substantial remediation.

G8

No Multi-Modal Fusion (Audio + Visual)

LOW PRIORITY

Emotion and authenticity analysis is performed exclusively on visual (image/video frame) data. Research consistently shows that combining facial expressions with vocal prosody (tone, pitch, speech rate) and physiological signals (rPPG heart rate from video) significantly improves emotion recognition accuracy. No audio stream analysis or rPPG-based liveness detection is implemented.

Impact: Emotion accuracy is ceiling-limited by unimodal visual input; multimodal fusion could improve F1-scores by 15–25% on benchmark datasets.

Research Gap Summary Matrix

#	GAP	AFFECTED MODULE(S)	PRIORITY	SUGGESTED RESOLUTION
G1	Demographic bias & dataset imbalance	Emotion, Age & Gender	High	FairFace fine-tuning, oversampling, bias audits
G2	Deepfake detection gap vs. diffusion models	Deepfake	High	Integrate CNN classifier trained on FF++ & DFDC
G3	Bruise detection has no real ML model	Bruise & Marks	High	Fine-tune YOLOv8 on annotated injury datasets
G4	No explainability / XAI layer	All modules	Medium	Integrate Grad-CAM, LIME, or attention visualizations
G5	No cross-session temporal tracking	All modules	Medium	Add persistent DB (PostgreSQL/Firebase) + user profiles
G6	Single-face & occlusion sensitivity	All modules	Medium	Multi-face pipeline, pose normalization, occlusion handling
G7	No privacy / consent / ethics framework	Whole platform	Medium	GDPR consent UI, anonymization, on-device inference option

#	GAP	AFFECTED MODULE(S)	PRIORITY	SUGGESTED RESOLUTION
G8	No multi-modal fusion (audio + visual)	Emotion, Deepfake	**Low**	Add audio emotion analysis + rPPG liveness detection

Human Authenticity Analyzer — Technical Architecture

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